

# evaluate\_functional\_type\_classification\_change.R

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| evaluate_functional_type_classification_change          |
| <i>Evaluate a Functional-Type Classification Change</i> |

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## Description

Compares two functional-type classifications by taxon, raw cluster labels, label-invariant co-clustering relationships, and silhouette widths. The partition comparison is invariant to arbitrary integer relabelling of otherwise identical clusters.

## Usage

```
evaluate_functional_type_classification_change(data_reference, data_candidate)
```

## Arguments

`data_reference` A data frame with unique ‘taxon\_name’ values and columns ‘functional\_type’ and ‘silhouette\_width’.

`data_candidate` A data frame with the same required columns as ‘data\_reference’.

## Details

Label-invariant partition differences are counted from cluster-size and contingency-table combinations. This avoids constructing a quadratic taxon-by-taxon matrix while counting pairs that are in the same cluster in exactly one classification.

## Value

A named list with ‘data\_summary’, a one-row tibble of comparison counts, and ‘data\_taxon\_comparison’, a taxon-level full join of both classifications.

**Examples**

```
data_reference <- tibble::tibble(  
  taxon_name = c("a", "b", "c"),  
  functional_type = c(1L, 1L, 2L),  
  silhouette_width = c(0.4, 0.5, 0.6)  
)  
data_candidate <- tibble::tibble(  
  taxon_name = c("a", "b", "c"),  
  functional_type = c(2L, 2L, 1L),  
  silhouette_width = c(0.4, 0.5, 0.6)  
)  
evaluate_functional_type_classification_change(  
  data_reference,  
  data_candidate  
)
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